Week 5 – Lists & Dictionaries

■ Learning Objectives

• Create, access, and modify Python lists.
• Use dictionaries for fast lookups and frequency counts.
• Apply loops to process collections and solve CCC-style tasks.

■ Key Concepts

1. Lists: index, slice, append, pop, remove, sort, reverse.
2. Iteration: for-each vs index-based loops; enumerate.
3. Dictionaries: key→value mapping, membership testing, get(), setdefault().
4. Patterns: counting frequency, deduplicate, grouping.

Example 1 – List Basics

```
nums = [5, 2, 9, 1, 5]
print(nums[0], nums[-1])
nums.append(7)
nums.remove(2)
nums.sort()            # in-place sort
print(nums)            # [1,5,5,7,9]
```

Example 2 – Frequency with Dict

```
words = "to be or not to be".split()
freq = {}
for w in words:
    freq[w] = freq.get(w, 0) + 1
print(freq)  # {'to':2, 'be':2, 'or':1, 'not':1}
```

■ Practice Problems

Practice 1 – Remove Duplicates (Keep Order)

Input a list of integers; output without duplicates, preserving order.

Solution:

```
data = list(map(int, input().split()))
seen = set()
out = []
for x in data:
    if x not in seen:
        out.append(x)
        seen.add(x)
print(*out)
```

Practice 2 – Most Frequent Word

Read a line, print the word with highest frequency (break ties lexicographically).

Solution:

```
words = input().split()
freq = {}
for w in words:
    freq[w] = freq.get(w, 0) + 1

best = None
for w in sorted(freq):  # ensures lexicographic tie-break
    if best is None or freq[w] > freq[best]:
        best = w
print(best)
```

Practice 3 – Pair Sum Finder

Given a list of numbers and target T, print indices of two numbers that add up to T (first pair).

Solution:

```
arr = list(map(int, input().split()))
T = int(input())
pos = {}
ans = None
for i, x in enumerate(arr):
    need = T - x
    if need in pos:
        ans = (pos[need], i)
        break
    pos[x] = i
print(ans if ans else "None")
```

■ Homework Challenge

Read N lines of 'name score' and print the name with the highest total score.

Expected Solution:

```
n = int(input())
tot = {}
for _ in range(n):
    name, s = input().split()
    s = int(s)
    tot[name] = tot.get(name, 0) + s
print(max(tot, key=tot.get))
```